GIS Test: Thought Process

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I did the GIS test in QGIS 3.44. It has been a while since I've used QGIS, so things took longer than I anticipated, and it took me a while to familiarize myself with how the filtering works. Getting familiar with the data while doing the Python assessment did help me navigate: I already knew DDOT's TOPS permit layers, the Roadway SubBlock layer and its `SUBBLOCKKEY`, and that DDOT's REST services accept SQL `WHERE` clauses. I used Claude to help me understand the data structure and help navigate QGIS.

My first instinct was to clean the data in Python first, the way I did in the Python test. Instead, I connected QGIS directly to DDOT's REST services and filtered each layer there, because the deliverable asks for the three feeds on the map and for the geoprocessing history. The filters are the same kind of SQL `WHERE` clause I used in Python; QGIS just sends them to the server.

For the permits, I kept renewals (`Renewal = 2`) that have been issued (`Status = 9`) and are excavation or paving permits (`IsExcavation = 1 OR IsPaving = 1`), expiring on or after October 1, 2025. Every record in the layer already expires after that date, so that condition removes nothing, but I kept it so the filter states the full requirement. I read "issued" as the Issued status, which leaves out permits that have since expired; in a work setting I would ask DDOT whether those should count. For the roadway, I kept road types 1, 2, and 3 (street, service road, and ramp). For Cityworks, I used DDOT's SQL as given. It keeps one request with the status "GEO ISSUE", which is not one of the listed statuses, and I kept it because the query came from DDOT. Learning the filtering was the slowest part: QGIS's "Add Filtered Layer" checks the text as a QGIS expression and rejects ArcGIS's `timestamp` syntax, so I added each layer with the date-free part of its filter and then entered the full SQL with Filter…, whose Test button confirmed the counts: 8,591 permit records, 23,104 subblocks, and 696 requests.

Two things about the data shaped the rest of the work. One permit record is one work location, not one permit: the 8,591 records belong to 2,024 permits, the same pattern as the occupancy permits in the Python test. And neither the permits nor the requests carry a subblock key, so each point had to be matched to its block by location.

After I got the hang of things, I made sure each feature was properly projected. I copied each filtered feed into a GeoPackage in EPSG:26985 (NAD83 / Maryland), which is in meters, so distances are real distances. My first copy went to a temporary layer by default, so I ran it again into the GeoPackage. Then I spatially joined the features, matching each permit and each request to its nearest street block. Every record matched exactly one block, but 901 permits and 112 requests sat more than 10 meters from a street block, because DDOT's system had placed them on alleys or driveways. To build the snapped permits, I moved each permit to the closest point on its matched

centerline, so all 8,591 now sit exactly on their block.

Claude Opus 5.5 caused an issue here. Following its instructions, the joined layers and the snapped permits came out without a coordinate system: the coordinates were right, but the layers no longer said they were in EPSG:26985, so QGIS or ArcGIS could not place them correctly. Its instructions did not include a check for this after the join. I fixed the snapped permits with the Assign projection tool, which relabels the layer without moving any points.

For the enrichment, I summarized every block in one Execute SQL step, since I am comfortable with SQL. The SQL took me a while, and I had Claude help me after a bit. It counts distinct permits on each block, because one permit can have several work locations on the same block, and keeps the number of locations too. It also counts the requests, takes the latest expiration and request dates, and flags blocks with permits, requests, and both. The GeoPackage stores dates in UTC, so I converted the request times to Eastern; otherwise a request made late in the evening would land on the next day.

Of the 23,104 street, service road, and ramp blocks, 2,568 have at least one permit, 585 have at least one open request, and 109 have both. The latest permit expires on October 5, 2027, and the latest open request was made on October 15, 2026, at 10:39 PM Eastern.

I find the current map style a bit of an eyesore. With more time, I would spend it enhancing the readability: show only the final layers, color the blocks by category (gray for neither, blue for permits only, orange for requests only, and a thick dark red for both), and show the permit and request points only when zoomed in.